SOE203 Software Engineering 1
Lab 1: Introduction

Suggested answers

1. What are the four important attributes that all professional software should have? Suggest four other attributes that may sometimes be significant.
   Answer
   Maintainability, dependability and security, efficiency, acceptability

   Correctness, reusability, portability, usability, interoperability

2. Develop your own answers to the five questions in the lecture note about software development challenges. Discuss them with your fellow classmates.
   Answer
   - Why does it take so long to get software finished?
     - Most software has a handful of user requirements, each requirement has to be carefully studied and understood. For some software engineering methods, documentation may take up a lot of effort. Requirement change prolongs the development schedule. A single change may affect many other areas of the software.

   - Why are development costs so high?
     - The development of a software requires a lot of skillful and experienced engineers. They are highly educated and expensive work force. The development of software requires not only to study the user requirements carefully, but also needs to design the solution and implement the design in term of software. After the implementation, the produced software has to be tested. As you can see, the development of software is a lengthy process, which spends expensive manpower.

   - Why can't we find all errors before we give the software to our customers?
     - Software evaluation techniques cannot spot all potential errors. Some errors occur only with the input of some specific data or specific situation where it might happen only in the production environment.
     - Errors in software also occur due to the change of production environment which is not possible to be simulated in the development environment, for example, a change of hardware.

   - Why do we spend so much time and effort maintaining existing programs?
     - Once a system has been in use for a significant period, replacing it becomes highly risky and costly. The system often embeds complex business logic, handles numerous edge cases, and supports established workflows that are not easily replicated. Therefore, when bugs arise or new requirements emerge, it is

almost always more practical and cost-effective to maintain and upgrade the existing system rather than attempt a full replacement. Maintenance preserves institutional knowledge, reduces user disruption, and avoids the unforeseen pitfalls of a ground-up rebuild.

- Why do we continue to have difficulty in measuring progress as software is being developed and maintained?
  - Unlike civil engineering, software engineer builds software which is intangible. It is hard to measure the progress of the virtual product which is still in progress. Unlike building a skyscraper, when you say you have built 20 floors, others can count and verify that you have 20 floors. In software development, sometimes the halfway done software cannot be verify because unfinished software cannot be executed. Even if you have a halfway done software, it may consist of serious bug which requires unknown amount of time to fix.

3. What is the most important difference between generic software product development and custom software development? What does this mean in practice for users of generic software products?

Answer

The essential difference is that in generic software product development, the specification is owned by the product developer. For custom product development, the specification is owned and controlled by the customer. The implications of this are significant: the developer can quickly decide to change the specification in response to some external change (e.g., a competing product) but, when the customer owns the specification, changes have to be negotiated between the customer and the developer and may have contractual implications.

For users of generic products, this means they have no control over the software specification so cannot control the evolution of the product. The developer may decide to include/exclude features and change the user interface. This could have implications on the user's business processes and add extra training costs when new versions of the system are installed. It may also limit the customer's flexibility to change their own business processes.

4. Many modern applications require modification frequently (before they are presented to the end user and then after the first version has been put into use). Suggest a few ways to build software to stop deterioration due to changes.

Answer

- The software should be modular so that changes will not have a lot of side effects to the other parts of the software.

- The software must be maintainable.
- Comprehensive testing should put in place to reduce errors.
- Work closely with the stakeholder(s) to ensure requirements are correctly defined.
- Improve the requirement study approach(es) to achieve a better requirements study.

5. Discuss whether professional software engineers should be certified in the same way as doctors or lawyers.

Answer

Yes, professional software engineers should be certified, much like doctors and lawyers, because the stakes are similarly high.

Unqualified doctors can cause physical harm or loss of life; unqualified lawyers can ruin livelihoods and freedom. The same principle applies to engineers. A civil engineer who designs a collapsing bridge or skyscraper endangers public safety. A software engineer who writes faulty code for financial systems can cause massive economic damage, and one who works on high-speed train controls could potentially cost hundreds of lives if the code is not rigorously designed and tested.

Certification ensures a baseline of competence, accountability, and ethical responsibility, which are protecting the public from avoidable disasters.

Advantages of certification for engineers:

- Certification raises the overall standard of the profession by ensuring engineers are trained according to recognized standards and legal requirements. It also provides a trusted, verifiable source of assurance that a certified engineer is competent and reliable.
- Certification enhances public trust and improves the reputation of the engineering profession, as it signals a commitment to ethics, quality, and accountability.
- For employers, certification offers a clear, objective benchmark of minimum competence, simplifying the screening process and helping them quickly identify candidates who meet essential professional standards.

It is important to note that certification does not guarantee competence but does guarantee that a minimum standard was reached at the time of the certification.